



IUS
INSTITUT
UNIVERSITAIRE
DES SCIENCES

Projet 4 : Implémentation des ACL pour le Contrôle d'Accès Réseau

Préparé par :

Dochlie Louis

Magdala Isidore

Alexandre Frantzdy

Sous la direction de: SAINT AMOUR Ismael

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Objectifs du Projet : Comprendre les Access Control Lists. Filtrer le trafic réseau. Améliorer la sécurité.

Plan du Projet : 1. Topologie réseau Créer réseau avec plusieurs VLAN.

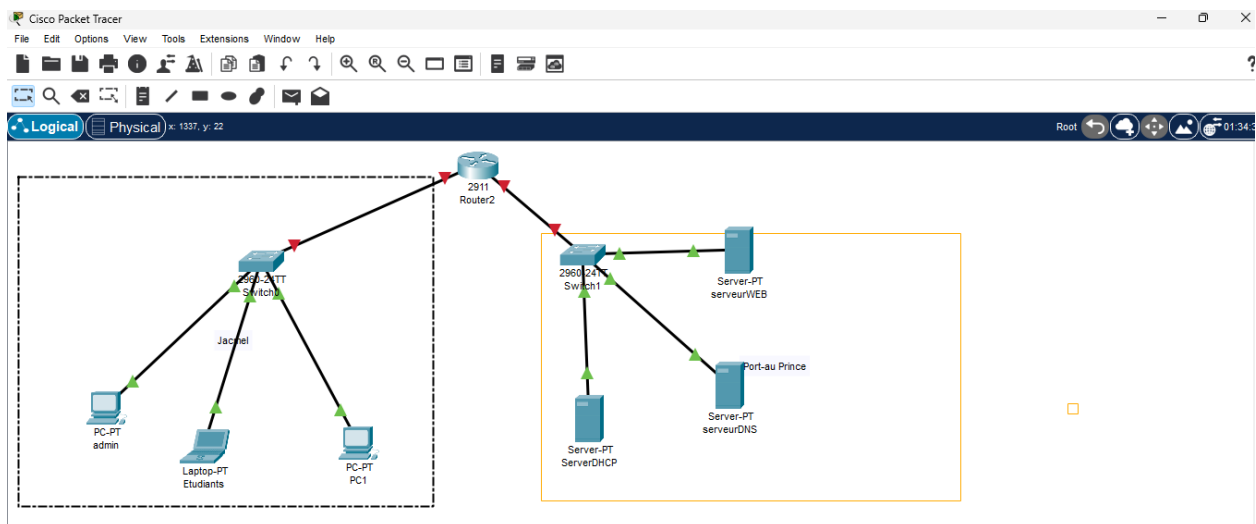
2. ACL standard Configurer ACL pour contrôler accès par IP.

3. ACL étendue Filtrer selon : protocole port

4. Application des ACL Appliquer sur interfaces routeur.

5. Tests Tester accès autorisés et refusés

Topologie du réseau



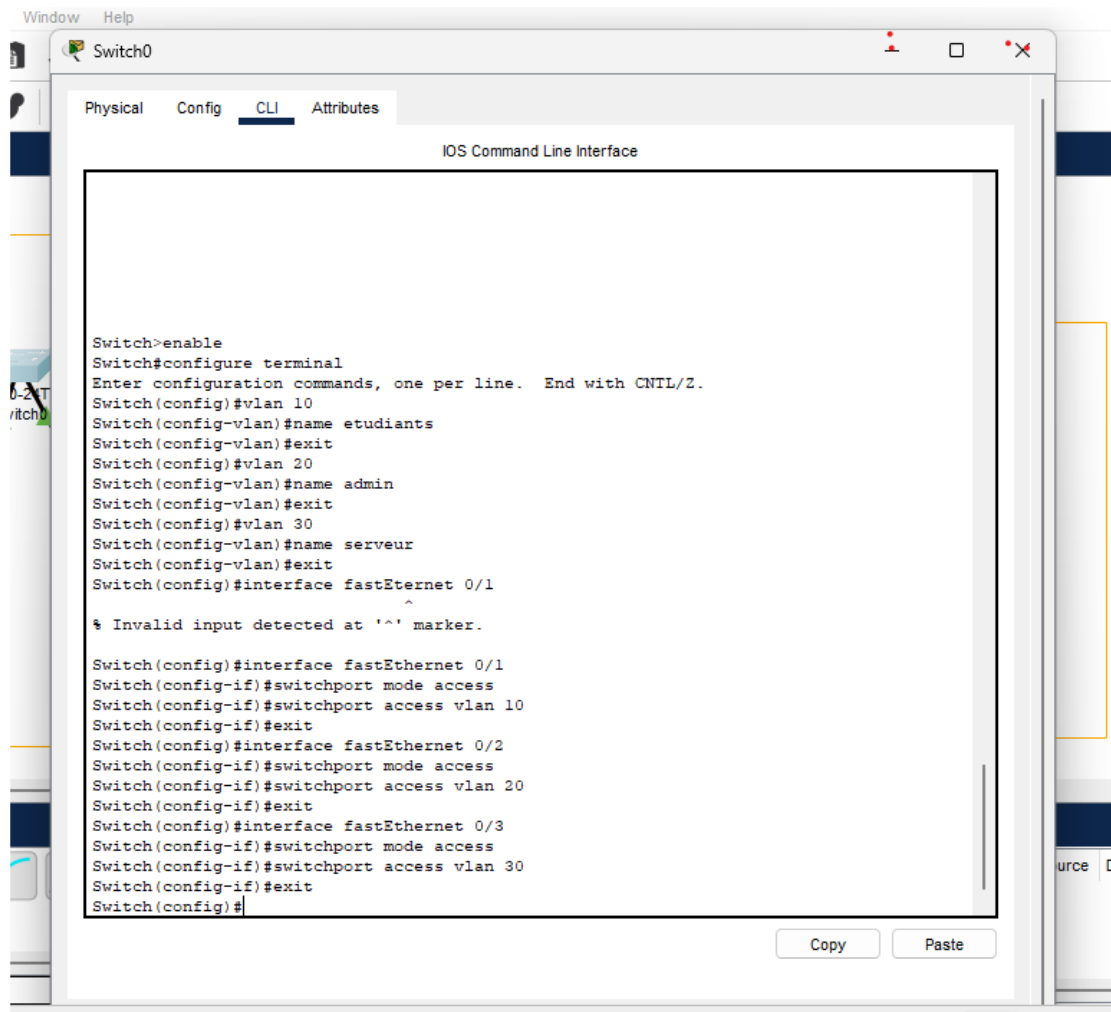
Configuration du switch (vlans)

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name etudiants
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name admin
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name serveur
Switch(config-vlan)#exit
Switch(config)#
```

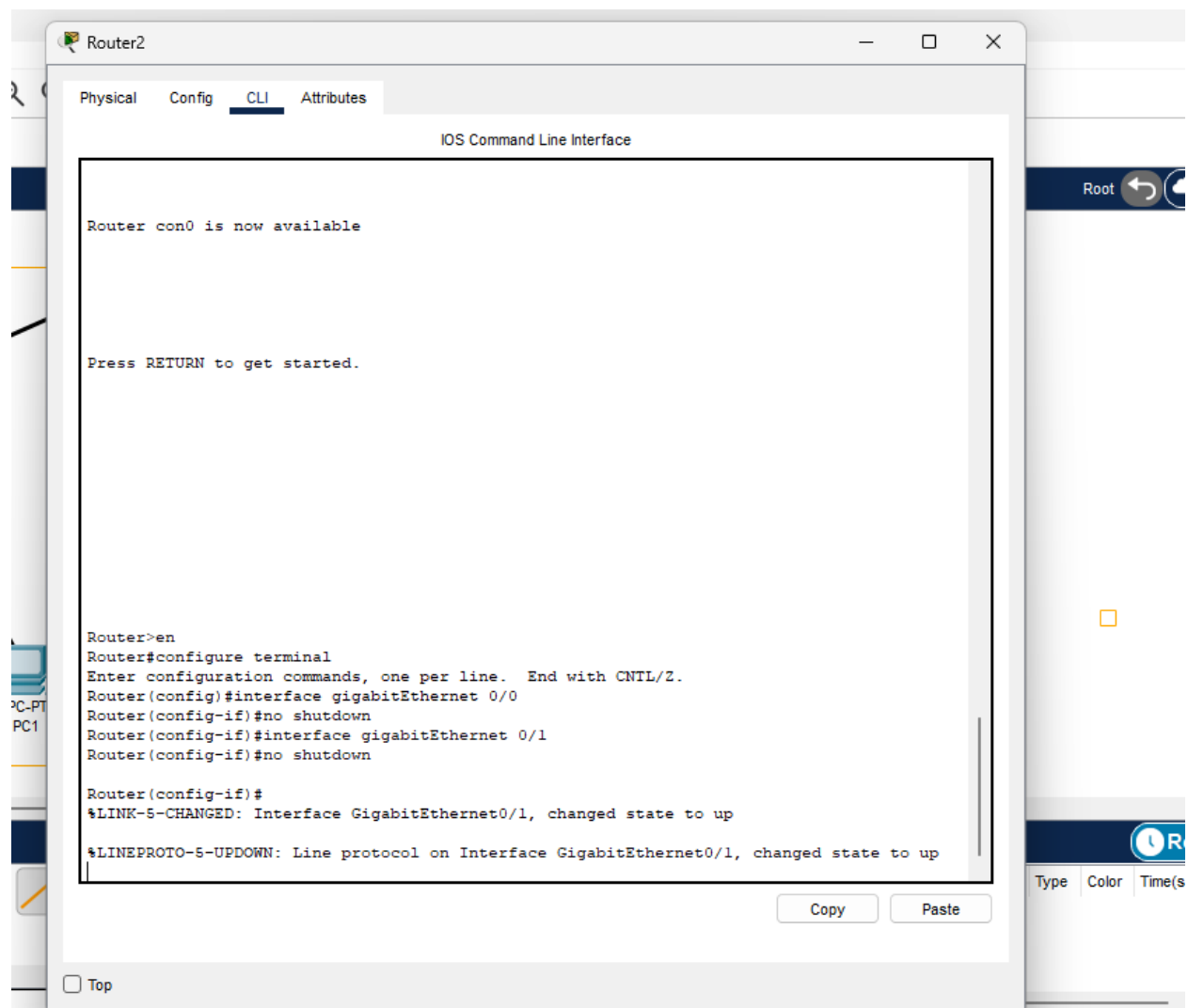
Copy

Paste

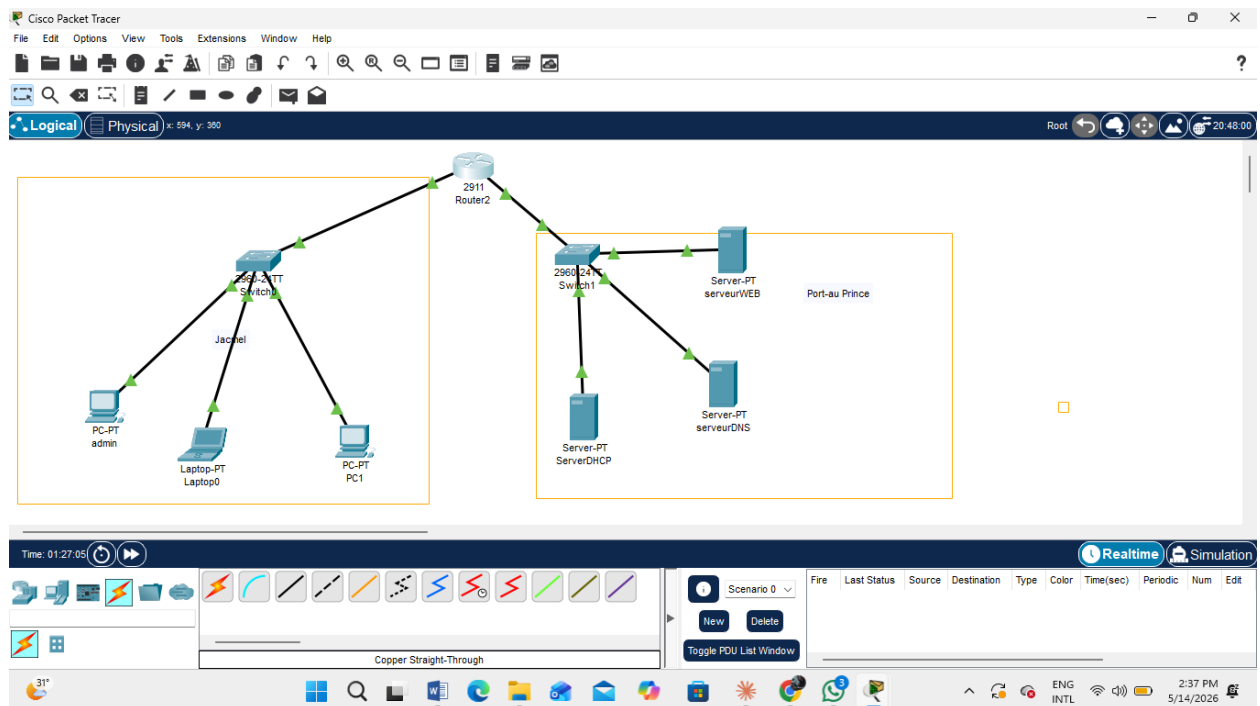
☐ Top



Configuration du routeur



Topologie avec le routeur et les sous interfaces configurés



admin

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.10.2

Subnet Mask

255.255.255.0

Default Gateway

192.168.10.1

DNS Server

0.0.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

Link Local Address

FE80::202:17FF:FEEA:E687

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

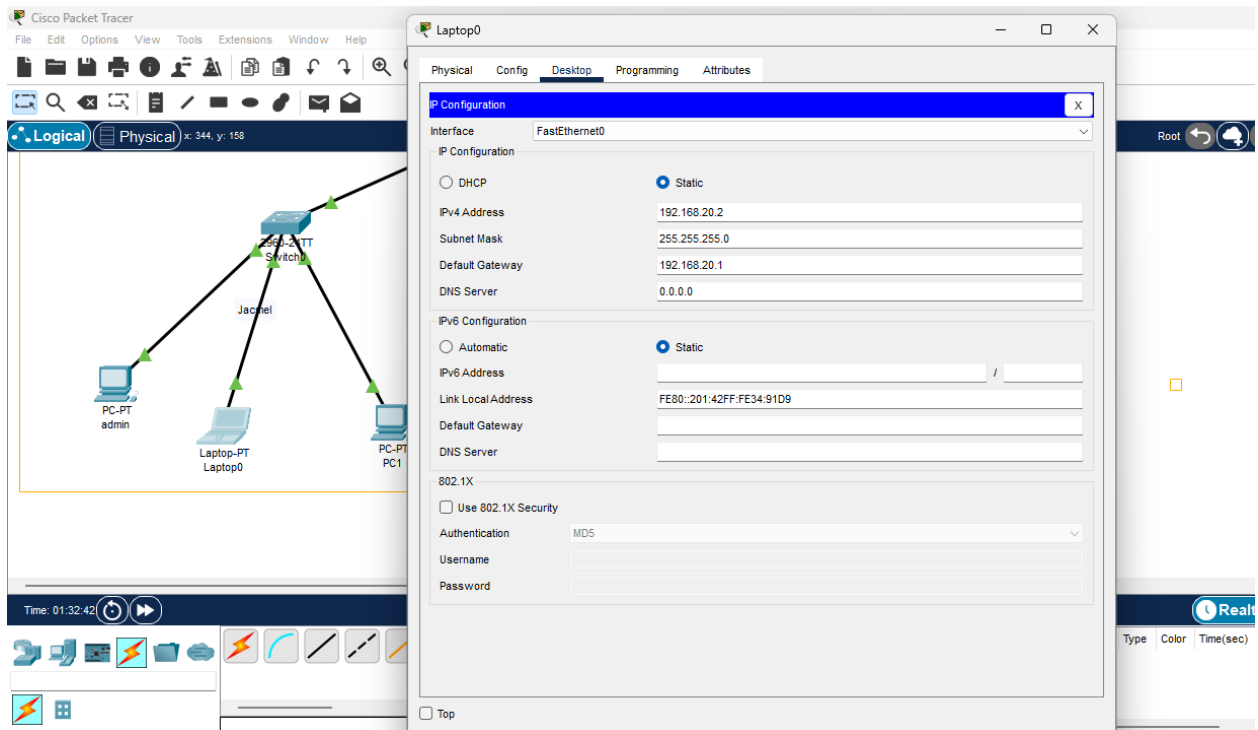
Authentication

MD5

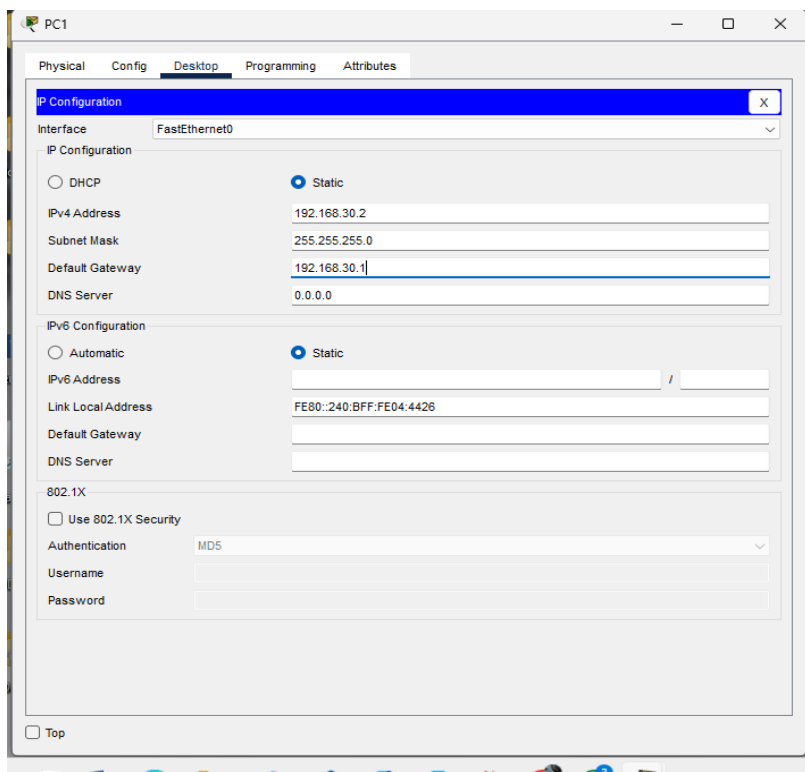
Username

Password

☐ Top



Configuration du Pc



Configuration du serveur DHCP

The screenshot displays the Cisco Packet Tracer interface. On the left, a network topology is visible in the 'Physical' tab, showing a central 'Switch' connected to three devices: 'PC-PT admin', 'Laptop-PT Laptop0', and 'PC-PT PC1'. The 'Switch' is labeled 'Jacquel'. The main window on the right is the 'ServerDHCP' configuration window, which is currently in the 'Desktop' tab. The 'IP Configuration' section is active, showing the following settings:

- ☐ DHCP
- ☒ Static
- IPv4 Address: 192.168.10.3
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1
- DNS Server: 0.0.0.0

The 'IPv6 Configuration' section is also visible, showing the following settings:

- ☐ Automatic
- ☒ Static
- IPv6 Address: /
- Link Local Address: FE80::201:C7FF:FEAB:54B7
- Default Gateway: /
- DNS Server: /

The '802.1X' section is also visible, showing the following settings:

- ☐ Use 802.1X Security
- Authentication: MDS
- Username: /
- Password: /

The bottom status bar shows the time as 01:41:07 and the simulation mode as 'Realtime'.

ServerDHCP

Physical

Config

Services

Desktop

Programming

Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DHCP

Interface

FastEthernet0

Service

On

Off

Pool Name

serverPool

Default Gateway

192.168.10.1

DNS Server

0.0.0.0

Start IP Address :

192

168

10

3

Subnet Mask:

255

255

255

0

Maximum Number of Users :

253

TFTP Server:

0.0.0.0

WLC Address:

0.0.0.0

Add

Save

Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168....	0.0.0.0	192.168....	255.255....	253	0.0.0.0	0.0.0.0

Top

Type

Configuration du serveur DNS

The screenshot shows a configuration window titled 'serveurDNS' with a tabbed interface. The 'Desktop' tab is selected, displaying three configuration sections: IP Configuration, IPv6 Configuration, and 802.1X.

IP Configuration

- ☐ DHCP
- ☒ Static
- IPv4 Address: 192.168.10.4
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1
- DNS Server: 0.0.0.0

IPv6 Configuration

- ☐ Automatic
- ☒ Static
- IPv6 Address: [empty] / [empty]
- Link Local Address: FE80::230:A3FF:FE9A:ABDD
- Default Gateway: [empty]
- DNS Server: [empty]

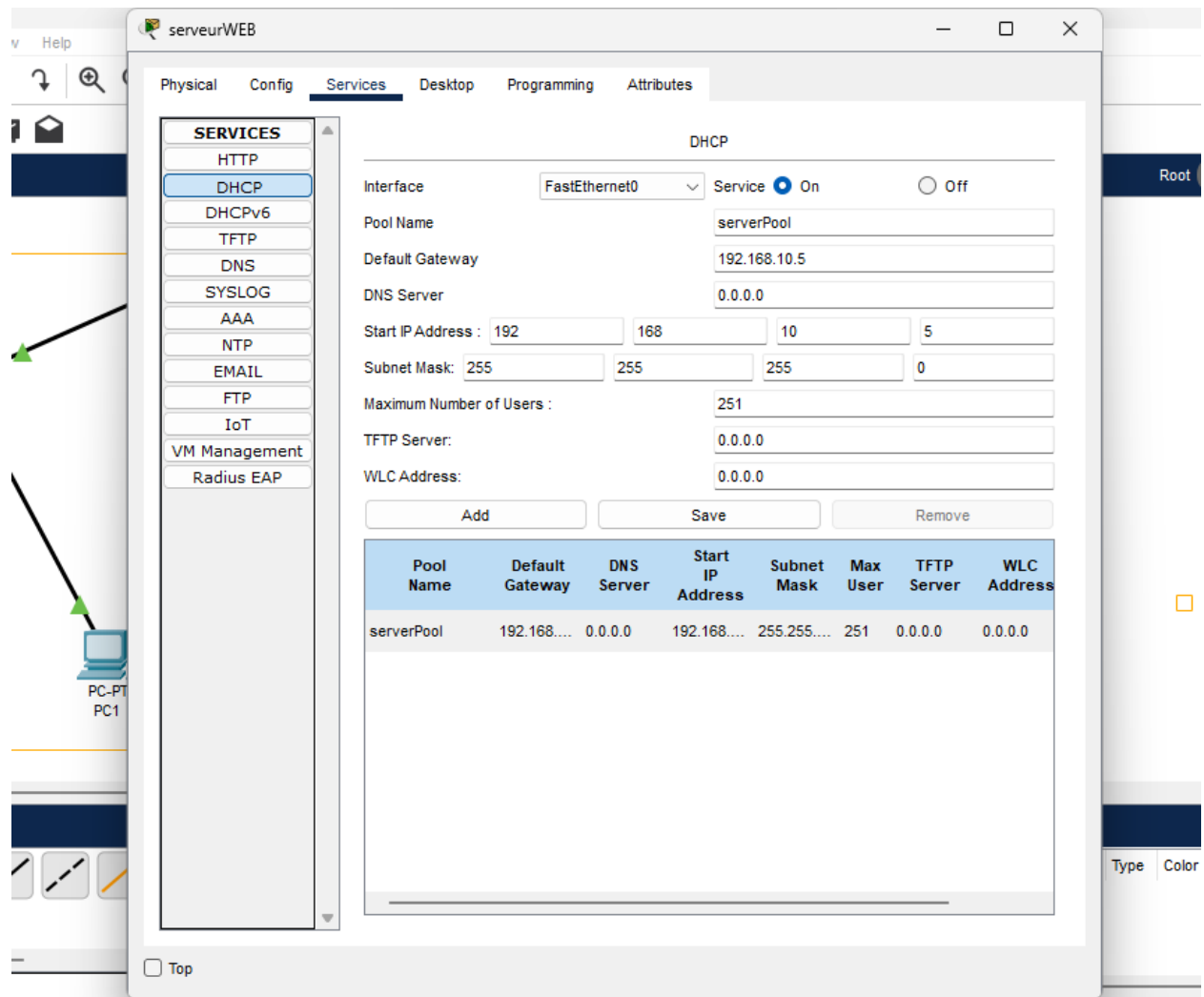
802.1X

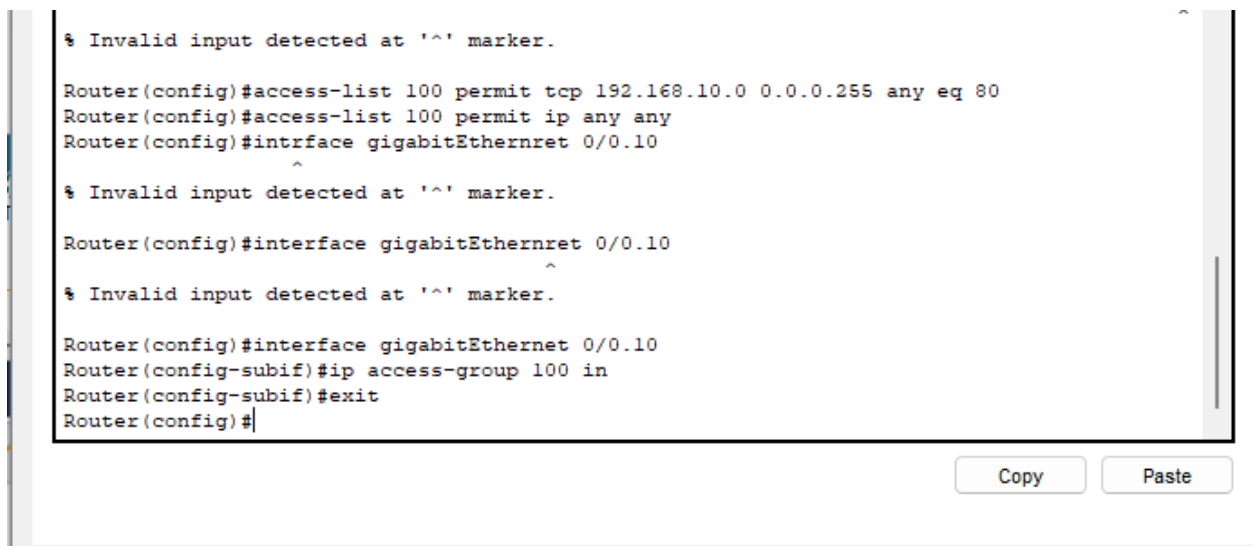
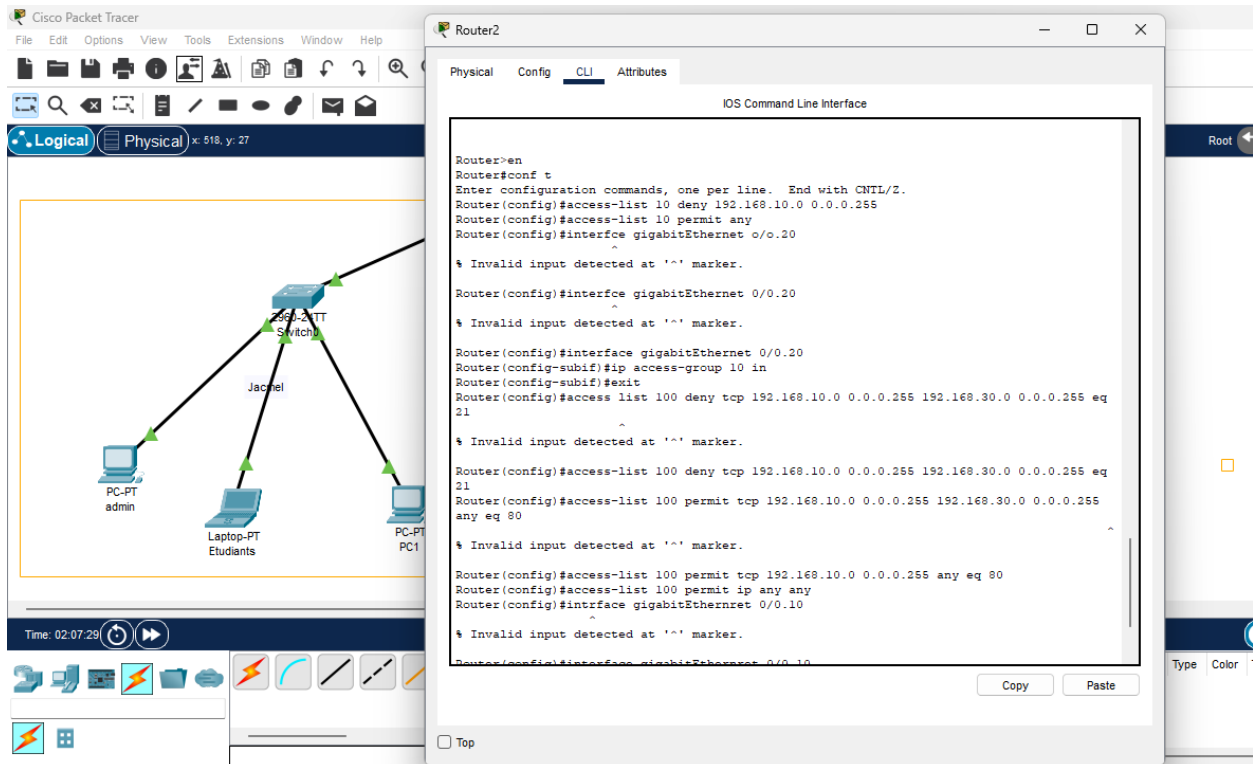
- ☐ Use 802.1X Security
- Authentication: MD5
- Username: [empty]
- Password: [empty]

☐ Top

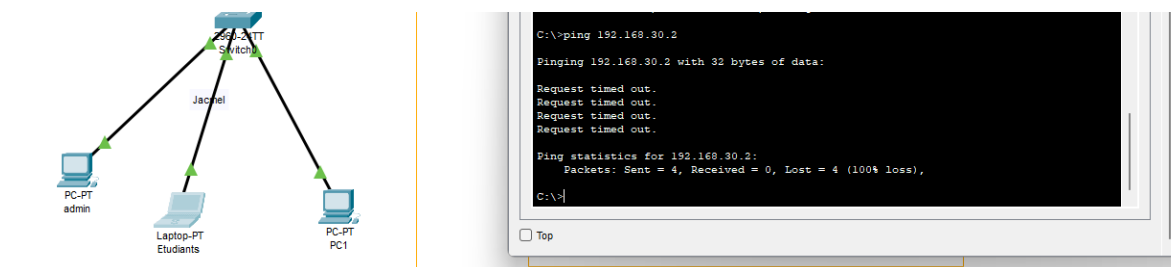
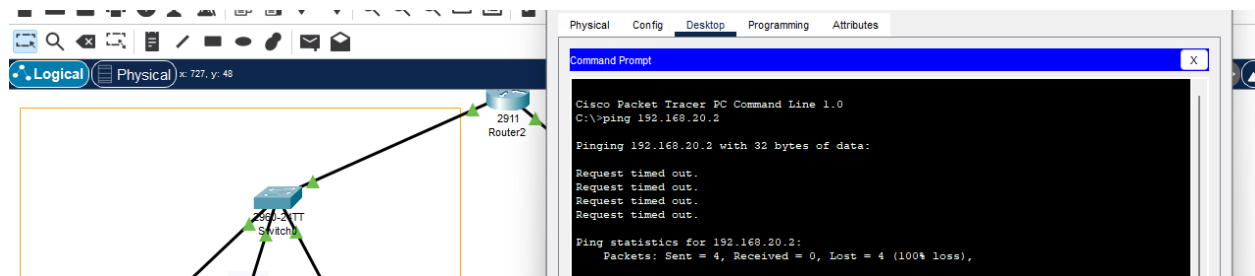
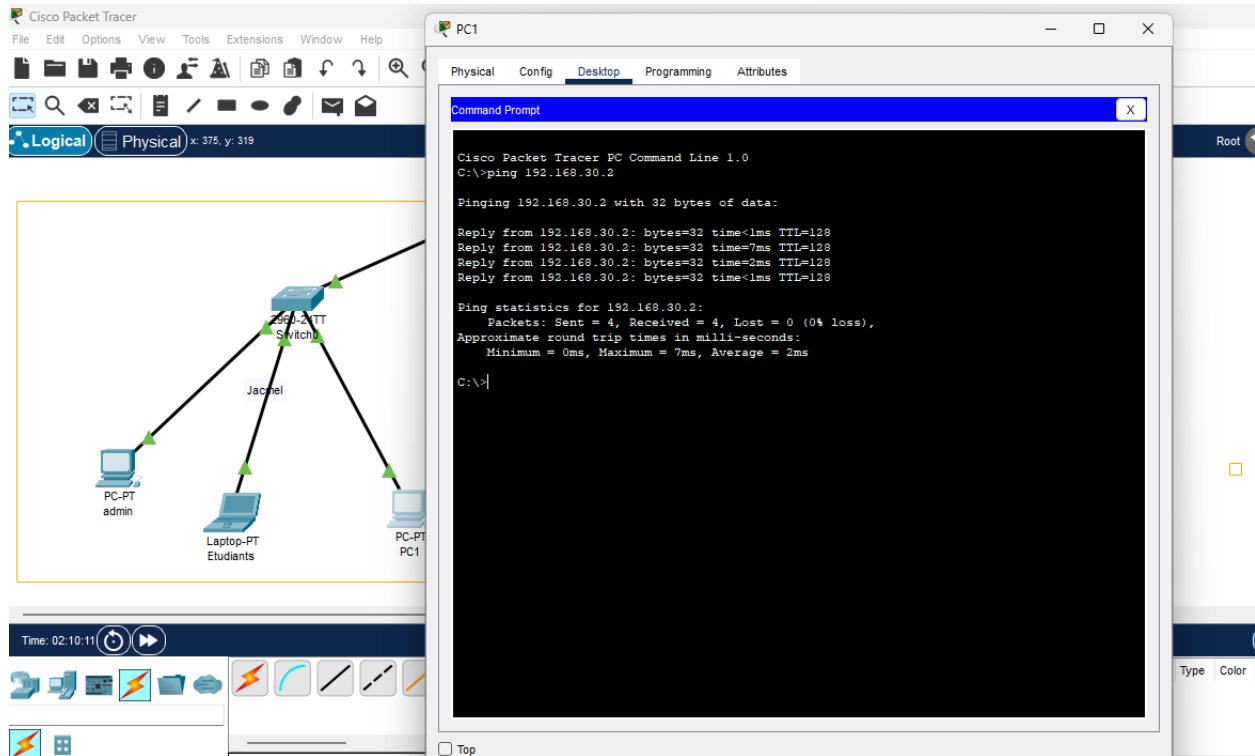
The screenshot shows the Cisco Packet Tracer interface with the 'serverDNS' configuration window open. The 'Services' tab is selected, and the DHCP configuration is shown. The DHCP pool is named 'serverPool', with a default gateway of 192.168.10.1 and a DNS server of 0.0.0.0. The IP address range is 192.168.10.1 to 192.168.10.252. The configuration is saved.

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168.10.1	0.0.0.0	192.168.10.1	255.255.255.0	252	0.0.0.0	0.0.0.0





Test et Résultats



Router2

Physical

Config

CLI

Attributes

IOS Command Line Interface

Press RETURN to get started.

Router>show vlan brief

VLAN	Name	Status	Ports
1	default	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>show access-list
      ^
% Invalid input detected at '^' marker.

Router>
Router>show access-lists
      ^
% Invalid input detected at '^' marker.

Router>do show access-lists
      ^
% Invalid input detected at '^' marker.

Router>
Router>do show ip access-lists
      ^
% Invalid input detected at '^' marker.

Router>do show ip access-list
      ^
% Invalid input detected at '^' marker.

Router> show ip access-list
      ^
% Invalid input detected at '^' marker.

Router>en
Router# show ip access-lists
Extended IP access list 101
 10 deny icmp any host 192.168.20.2
 20 deny tcp any host 192.168.30.2 eq ftp
 30 permit tcp any host 192.168.30.2 eq www
 40 permit icmp any host 192.168.30.2
 50 permit ip any any

Router#
```

Router2

Physical

Config

CLI

Attributes

IOS Command Line Interface

% Invalid input detected at '^' marker.

Router>en

Router# show ip access-lists

Extended IP access list 101

10 deny icmp any host 192.168.20.2

20 deny tcp any host 192.168.30.2 eq ftp

30 permit tcp any host 192.168.30.2 eq www

40 permit icmp any host 192.168.30.2

50 permit ip any any

Router#en

Router#show running-config

Building configuration...

Current configuration : 962 bytes

!

version 15.1

no service timestamps log datetime msec

no service timestamps debug datetime msec

no service password-encryption

!

hostname Router

!

!

!

!

!

!

no ip cef

no ipv6 cef

!

!

--More--

Copy

Paste

Type

C